

- 8 During radioactive decay, which is spontaneous and random, γ -ray (gamma ray) photons may be emitted. When these photons are incident on a sodium iodide crystal, some of the photons may be absorbed in the crystal. The absorption of a γ -ray photon causes the emission of a short pulse of light known as a scintillation. The scintillations may be detected and converted into electrical pulses using a *photomultiplier tube*, which, when connected to a counter, gives the *count rate* and enables γ -ray activity to be measured. The arrangement is illustrated in Fig. 8.1.

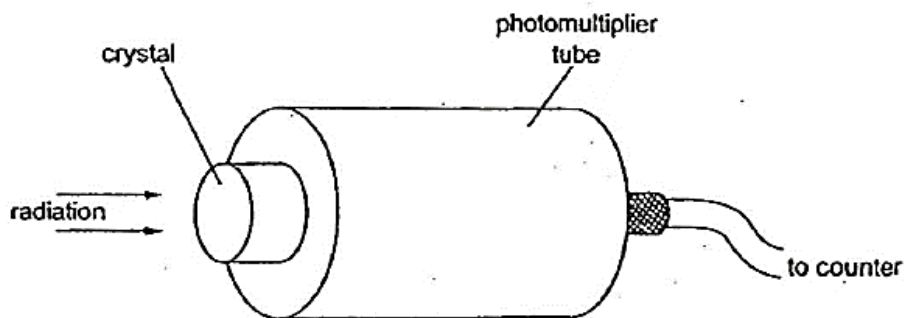


Fig. 8.1

The crystals used in such counters may be of different shapes. Fig. 8.2 shows a solid cylindrical crystal.

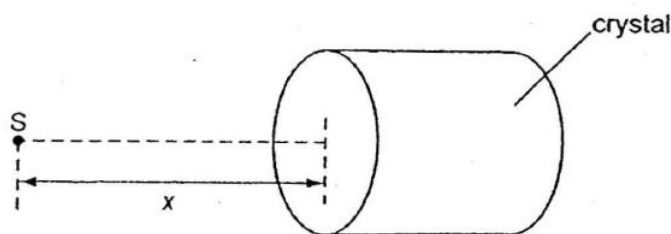


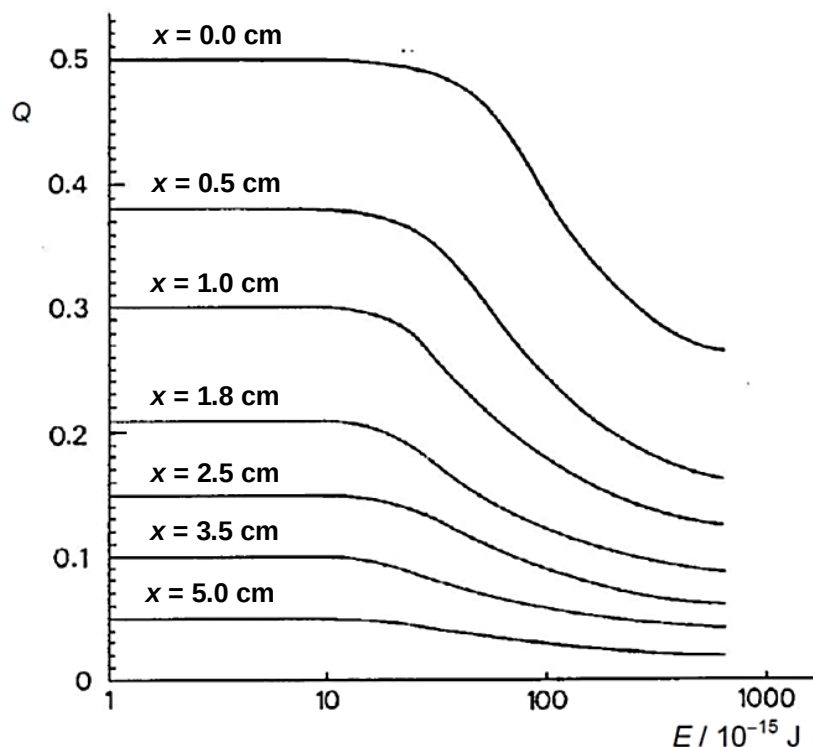
Fig. 8.2

The small γ -ray source S is placed a distance x in front of one face of the crystal. The source is assumed to emit photons uniformly in all directions. Not all of the emitted photons will be absorbed by the crystal. The efficiency Q of detection is defined by the equation

$$Q = \frac{\text{number of photons producing scintillations in the crystal}}{\text{total number of photons emitted by the source}}$$

(a)	By reference to the passage, explain what is meant by	
	(i) <i>count rate</i>	
	Count rate is a measure of the rate of radiation received by a radioactivity detector	
		
		

				
				[1]
		(ii)	activity	
			Activity of a radioactive isotope is defined as the number of nuclear disintegrations per unit time.	
				
				
				
				[1]
	(b)	Suggest two reasons why the γ -ray photons emitted by the source are not all absorbed in the crystal.		
			The source emits photons in all directions and many will travel in directions away from the face of the crystal and not be absorbed. Only those photons that arrive at the small circular window area of the face of the crystal will have a probability of being absorbed.	
		1.		
				
				
				[1]
			γ -rays are highly penetrating hence not every γ -ray photon that enters the crystal will interact with a sodium iodide molecule. Some of the higher energy photons will penetrate through the crystal without any interaction taking place.	
		2.		
				
				
				[1]
	(c)	Fig. 8.3 shows the variation with γ -ray photon energy E of the efficiency Q . Curves are drawn for various values of the distance x of the source S from the face of the crystal.		

**Fig. 8.3**

(i)	Suggest why at any one particular value of energy, the efficiency Q decreases as x increases.
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Intensity $I = P/A = P / 4\pi x^2$ where P is the power emitted by the radioactive source which is a constant.

As x increases, the photons are spread out over a larger spherical surface area A , and the intensity of the radiation incident on the face of the crystal will be lower.

A lower intensity means fewer photons arriving per unit area per unit time which will give a smaller Q value.

(ii)	With reference to Fig. 8.3 and considering γ -ray photons of energy 8×10^{-15} J, complete Fig. 8.4 with corresponding values of Q with for γ -ray photons of this energy.
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				Q	x / cm		
			Fig. 8.4				
							[2]
				Q	x / cm		
				0.05	5.0		
				0.10	3.5		
				0.15	2.5		
				0.21	1.8		
				0.30	1.0		
				0.38	0.5		
				0.50	0.0		

	(d)	(i)	Use your values in Fig. 8.4 to draw a graph of Q against x in Fig. 8.5.
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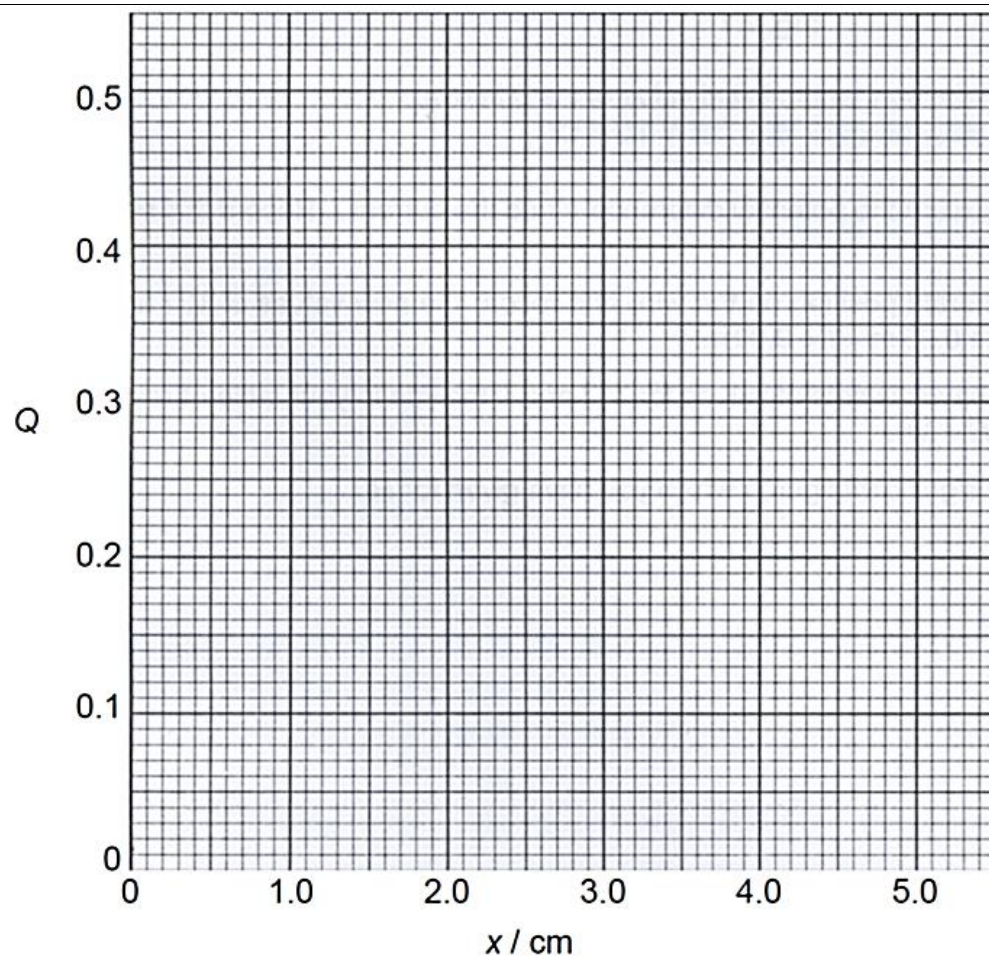
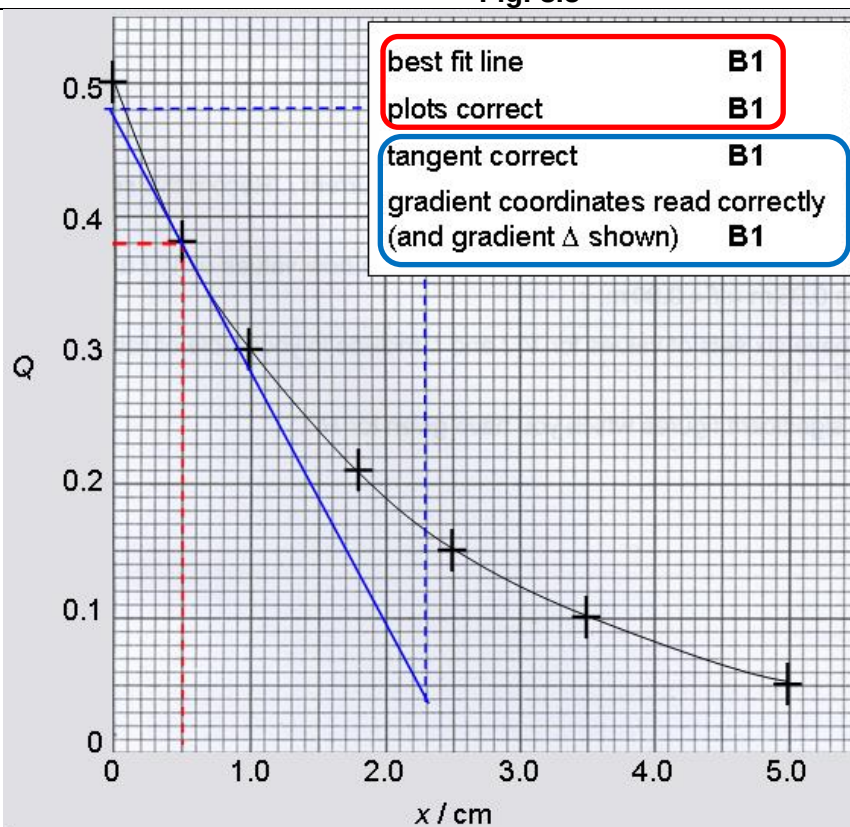


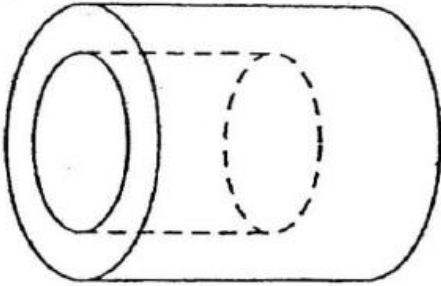
Fig. 8.5



[2]

[illegible]

	(e)	(i)	Suggest why the maximum efficiency that can be achieved with low-energy γ -ray photons using the crystal illustrated in Fig. 8.2 is 0.50.
			The maximum efficiency is achieved when $x = 0$ cm, i.e. the source is placed right at the face of the crystal. In this case, half the γ-ray photons emitted (those to the right side of the source within a hemispherical surface area) will be captured by the crystal.
			

				
				
				[2]
		(ii)	A second crystal consists of a hollow cylinder, as shown in Fig. 8.6.  Fig. 8.6	
			State the effect of this change of shape on the maximum efficiency Q of the detector if the source is placed inside the hollow region.	
			The maximum efficiency Q will be greater as the available area for capturing the γ -ray photons is larger.	
				
				[1]

END OF PAPER